Empirical Likelihood-Based Inference for Nonlinear Diffusions

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Abstract

Nonparametric methods are becoming popular recently in estimating continuous-time diffusion models due to their flexibility. This paper provides a new approach to constructing point-wise confidence intervals for nonparametric drift and diffusion functions via empirical likelihood (EL) in conjunction with local linear smoothing. It serves as a promising competitor to the conventional asymptotic normality-based method, which generally leads to (often seriously) under-covered confidence intervals in our simulation experiments, especially for the drift function. Limit theories are developed for the EL ratios for both drift and diffusion functions by means of increasing time span and shrinking observational intervals. Our results apply to both stationary and nonstationary recurrent diffusion processes. Compared to the asymptotic normality-based method, our approach avoids the difficulty of variance estimation by choosing the variance estimator implicitly as an operational nature of EL, and it has the additional advantage of automatic determination of the shape of the resultant confidence interval. Simulation studies show that for the both drift and diffusion functions, the EL confidence intervals are very accurate and robust to bandwidth choices even with moderate sample sizes, and are generally shorter (after confidence level calibration) than those based on asymptotic normality.

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1 Introduction

This paper develops methods of inference for the fully nonparametric and potentially nonstationary continuous-time diffusion model. Stochastic diffusion models have been widely used for describing the dynamics of underlying economic variables such as interest rates, exchange rates, asset prices, *etc.* Their successful applications in finance have been long demonstrated, see Merton (1992) and more recently, Sundaresan (2000) (see also Bergstrom, 1990, for other applications in macroeconomics). Estimation and inference in such models (as X_t in (1) below), involving the infinitesimal expected return (drift) and volatility (diffusion) functions is an important area in modern econometrics.

When parametric forms of the two infinitesimal moment functions are specified, several estimating methods and the associated inference procedures, such as the ones based on (exact or approximate) maximum likelihood, the generalized method of moments and simulations among others have been suggested. Recent surveys in this area are Sørensen (2004), Phillips and Yu (2005) and Aït-Sahalia (2006), to mention only a few. However, economic or financial theories provide little guidance on the choice of an appropriate parametric form of the model, leading to selection criteria that are often based on convenient mathematical derivations and ease of inference. Such parametric approaches work well when the true model is close to the specified one, but they are generally subject to potential (often significant) mis-specification errors. Accordingly, nonparametric methods have recently been developed for continuous-time diffusion models which impose no restrictive functional forms on the drift and diffusion functions beyond some regularity conditions. Compared to their parametric analogs, these approaches require much less prior information on the underlying data generating process and the resulting estimation and inference procedures are generally robust to model specifications. Nonparametric modeling is especially important in view of recent substantial evidence against most existing parametric models (see, *inter alia*, Aït-Sahalia, 1996b, Fan and Zhang, 2003, Hong and Li, 2005 for specification testing of short rate models), which raise concerns about

misleading economic implications.

In an effort to pricing interest rate derivative securities, Aït-Sahalia (1996a) proposed a semiparametric estimation method for continuous-time diffusion models. He assumed a linear mean-reverting specification of the drift function and the volatility term was estimated nonparametrically by matching the stationary density of the underlying process. See also Kristensen (2004) for more general semi-parametric diffusion modeling. A fully nonparametric estimation method for diffusions was proposed by Jiang and Knight (1997). They first used a kernel estimator of the diffusion function as in Florens-Zmirou (1993) and then estimated the drift function via a similar density-matching procedure to Aït-Sahalia's. Both the two-step procedures of Aït-Sahalia and Jiang and Knight depend on cross-restrictions of the drift and diffusion functions through the time-invariant density of the underlying process, so that the behavior of the first-step estimator usually affects the performance of the second-step estimator. Bandi and Phillips (2003) (BP hereafter) suggested a kernel-based functional estimation approach, nesting Stanton's (1997) estimator as a special case, to identify and estimate the drift and diffusion functions separately, considering the other function as a nuisance parameter. The intuition behind this approach is that the drift and diffusion functions are of different orders. Employing the local time concept to characterize the convergence rates, BP established limit theories for the drift and diffusion estimators by increasing both the time span and the sampling frequency (*viz.* long span and in-fill asymptotics). Their results do not rely on the existence of a stationary distribution and apply to a more general family of diffusion processes.

Stanton's and BP's methods are based on the Nadaraya-Watson or local level techniques, which are further extended by Fan and Zhang (2003) and Moloche (2001) by considering local linear fitting. Local linear estimators enjoy appealing bias properties as well as other theoretical advantages and are popular choices among practitioners. Nicolau (2003) and Fan and Yao (1998) considered discretization bias corrections of the local level and local linear volatility functional estimators, respectively. An inherent disadvantage of the local linear smoothing is that it may produce negative estimates for the volatility function, and in this case the reweighted functional estimator can be used (Xu, 2006). Sam and Jiang (2006) studied bias corrections in the nonparametric estimator of the short rate diffusion using a panel of yields

with different maturities. Finally, Cai and Hong (2003), Fan (2005) and Aït-Sahalia (2006) provide reviews of recent developments in nonparametric methods for continuous-time models.

In contrast to the large literature mentioned above providing nonparametric point estimators of the drift and diffusion functions, this paper focuses on the construction of (point-wise) confidence intervals. Confidence intervals are important in many empirical contexts where it is more useful to obtain a range of values with a high confidence level rather than just a single point estimate in inference. Point-wise confidence intervals suggest directions of variability of the underlying estimator, which is particularly relevant when the interval is not necessarily symmetric, and also provides a basis for the construction of a uniform confidence band to infer the shape of the true curve. The present paper confines attention to confidence interval construction around the local linear estimator for each of the two infinitesimal moment functions (*i.e.* the drift and diffusion), although the method provided here applies to other local polynomial (*e.g.* Nadaraya-Watson) techniques, as well as the nonparametric estimator of the stationary density function or the local time process in nonstationary cases.

The conventional method of constructing confidence intervals for the drift and diffusion functions is based on the asymptotic mixed normality of the corresponding nonparametric estimator. This entails estimation of the random variance terms, which is difficult even with large sample sizes. Different bandwidth choices for the underlying estimator and its variance estimator are needed in this construction. More importantly, the asymptotic distribution is a poor approximation in many cases, and the associated confidence intervals can suffer from severe under-coverage and high sensitivity to bandwidth choices, resulting in unreliable inferences. This is particularly true at points where the stochastic process visits sparsely and the information content in the data about the true quantities of interest is therefore slim.

To tackle these difficulties, our approach forms confidence intervals via empirical likelihood (EL), which was first introduced by Owen (1988, 1990) in the simple context of inference about the mean of an unknown distribution. Empirical likelihood can also be used through estimating equations. In our nonparametric dependent diffusion estimation context, we construct the EL ratio function through the estimating equation satisfied by the point-wise local linear estimator for the drift or diffusion function. Employing the long span and in-fill asymptotics

as in Bandi and Phillips (2003) we validate the Wilks-type limit theory (*i.e.* the chi-squared distributional limit of the log EL ratio), thereby delivering point-wise confidence intervals for the two infinitesimal moment functions. Our results apply to both stationary and non-stationary recurrent diffusion processes, thus accommodating the large empirical literature where martingale behavior is present in the data.

Compared to the conventional normal method, EL has the advantage of choosing the variance estimator implicitly for the local linear smoother, thereby avoiding secondary plug-in variance estimation. Multiple bandwidth choices are not needed. Additionally, the EL confidence interval has a data-determined shape unlike the normal confidence interval that is artificially centered at the corresponding local linear estimator. Our simulation studies show that for both drift and diffusion functions, the EL confidence intervals are very accurate (with very close empirical coverage probabilities to the nominal confidence level) even with weekly sampled data and are generally shorter (after confidence level calibration) than those based on asymptotic normality.

The current work extends the earlier literature on empirical likelihood methods combined with nonparametric regression. In the independent and identically distributed setting, Hall and Owen (1993) and Chen (1996) studied the EL confidence interval for a density function around its kernel estimate. In a closely related paper, Chen and Qin (2000) proposed the EL confidence interval for the mean function based on its local linear estimate and demonstrated several advantages of the EL procedure over that based on asymptotic normality. Their approach was further extended by Chan *et al* (2006) and Qin and Tsao (2005) to consider the volatility function and the derivative of the mean function respectively. As we will show below, Chen and Qin's method adapts nicely to the nonparametric continuous-time diffusion inference context, although a different characterization of the convergence rate is called for. For example, for the drift function, the point-wise confidence interval will become shorter only when the sample size increases in ways that expand the local time spent around the state point; in the case of positive recurrent and stationary diffusions, this means that the time span should be lengthened. Merely sampling more and more frequently over a fixed interval will not on its own make the confidence set for the drift function more concentrated. This reflects the fact established in BP that the

drift component can not be identified over a fixed time span without stationarity-based cross restrictions no matter how finely the process is sampled (see also Merton, 1980, Bandi, 2002, Fan and Zhang, 2003).

Empirical likelihood in conjunction with nonparametric regression has also been used in testing model specification. Based on a comparison of parametric regression estimators with their kernel smoothing analogs, Tripathi and Kitamura (2003) considered a smoothed EL-based test for a conditional moment restriction with independent and identically distributed observations (see also Chen, Härdle and Li, 2003). In a continuous-time diffusion context, Chen and Gao (2005) proposed an EL-based specification test assuming the existence of a time-invariant measure.

The remainder of the paper is organized as follows. Section 2 introduces the nonparametric diffusion model and outlines the assumptions imposed throughout the paper. This section also reviews the local time concept and the local linear estimators of the drift and diffusion functions, which are our main interests. In Section 3.1, we motivate the empirical likelihood procedure of forming confidence intervals for the drift function, and establish the large sample properties of the EL ratio statistics. The corresponding method of constructing confidence intervals for the diffusion function is given in Section 3.2. In Section 4 we compare the finite sample performance of our proposed inference procedures with the conventional confidence interval construction method based on asymptotic normality in a numerical example. Section 5 concludes. Proofs of the main results are gathered in a mathematical appendix.

2 The Model, Assumptions and Local Polynomial Estimation

2.1 The Diffusion model and assumptions

Suppose that we observe a discrete sample $\{X_{i\Delta}, i = 1, \cdots, n\}$ at n equally spaced time points from a continuous-time scalar Itô diffusion process $\{X_t, 0 \leq t \leq T\}$ satisfying the stochastic differential equation (SDE)

$$dX_t = \mu(X_t)dt + \sigma(X_t)dB_t, \tag{1}$$

where $\{B_t, 0 \leq t \leq T\}$ is a standard Brownian motion. Here Δ is the step between observations and T is the time span of the sample satisfying $T = n\Delta$. The nonrandom functions $\mu(\cdot)$ and $\sigma^2(\cdot)$ are called the infinitesimal mean function (or drift function) and infinitesimal variance function (or diffusion function, volatility function), respectively. We assume time-homogeneity of the diffusion process in the sense that two infinitesimal moment functions depend only on the current state, not on time. The diffusion process $\{X_t\}$ in (1) satisfies the (strong) Markov property and is a simplified version of the Brownian semi-martingale model where the drift and diffusion terms are general stochastic processes. The reader can think of $\{X_t\}$ as generated as follows: given the current state X_t, for a "tiny" increment in t, δt, the increment in the process X_t, $X_{t+\delta t} - X_t$, is a $(\mu(X_t), \sigma^2(X_t))-$Brownian motion over $(0, \delta t)$; the diffusion is then built up from Brownian motions continually re-adjusting its drift and variance according to the current state. The continuous-time diffusion in (1) contains many widely-used one-factor (parametric) financial models, such as the geometric Brownian motion (Osborne, 1959, Black and Scholes, 1973) for modeling stock price and several popular short-term interest rates models in the literature (for a partial list, see, *e.g.*, Table 1 in Aït-Sahalia, 1996a). Continuous-time modeling elegantly accommodates the fact that news usually arrives at financial markets in a continuous manner and provides a convenient framework for asset valuation and risk management.

In this paper, instead of imposing parametric restrictions on the two infinitesimal moment functions, we follow Bandi and Phillips (2003) by assuming the general conditions ensuring existence and pathwise uniqueness of a strong solution $\{X_t\}$ to the stochastic differential equation (1).

Assumption E (Existence). *Functions $\mu(\cdot)$ and $\sigma(\cdot)$ are measurable on $\mathcal{D} = (l, u)$ with $-\infty \leq l < u \leq \infty$. They are at least twice continuously differentiable and satisfy Lipschitz and growth conditions, viz. for any compact subset $S \subseteq \mathcal{D}$, there exists constant M_S and M'_S such that for all $x, y \in S$, $|\mu(x) - \mu(y)| + |\sigma(x) - \sigma(y)| \leq M_S|x - y|$ and $|\mu(x)| + |\sigma(x)| \leq M'_S(1 + |x|)$. Furthermore, $\sigma^2(\cdot) > 0$ on $\mathcal{D}$.*

Define the scale function of the diffusion process $\{X_t\}$ as $S(\alpha) = \int_c^\alpha \exp\left(\int_c^y \left(-\frac{2\mu(x)}{\sigma^2(x)}\right) dx\right) dy$, $c \in \mathcal{D}$. The following condition guarantees that $\{X_t\}$ is recurrent, which intuitively means the process will visit every state $x \in \mathcal{D}$ infinite times with probability one.

Assumption R (Recurrence). $\lim_{\alpha \to l} S(\alpha) = -\infty$ and $\lim_{\alpha \to u} S(\alpha) = \infty$.

For a Markov stochastic process, recurrence is implied by positive recurrence (or ergodicity) which assumes the existence of a stationary distribution. If a positive recurrent process starts with its stationary distribution, it is strictly stationary. In other words, recurrence is a more general specification of a Markov process than positive recurrence or stationarity. Bandi and Phillips (2002, 2003) show that the recurrence (including both null recurrence and positive recurrence) condition is sufficient to identify the drift and diffusion function from a discrete sample. This is an important extension in view of recent empirical studies of U.S. interest rates dynamics which reveal that the estimated drift term is actually very close to zero (Bandi, 2002, Durham, 2003, Fan and Zhang, 2003). In these cases, recurrence is a reasonably weak assumption that allows for such martingale behavior.

Nonparametric estimation of more complicated models than (1) have also been studied, *e.g.* jump diffusions (Bandi and Nguyen, 2003, Johannes, 2004), multivariate diffusions (Bandi and Moloche, 2001) and stochastic volatility models (Reno, 2006). In particular, substantial recent attention has been given to robust estimation with high frequency observations contaminated by the market microstructure noise. The inference procedures suggested in this paper should provide a theoretical basis to analyze such extended models.

2.2 The Local time and its estimator

The local time concept plays a key role in estimating recurrent diffusions. The local time (chronological local time, sojourn time) of the diffusion process (1) is defined as

$$\overline{L}_X(T, x) = \lim_{\varepsilon \to 0} \frac{1}{\sigma^2(x)\varepsilon} \int_0^T 1_{(x, x+\varepsilon)}(X_s)\sigma^2(X_s) ds, \ \forall x \in \mathcal{D}, T \geq 0. \tag{2}$$

This random quantity, depending on the time span T and the state point x, can be interpreted as the amount of calendar time spent by the diffusion process $\{X_t\}$ in the neighborhood of x up to time T. The local time is a useful descriptive tool in the analysis of nonstationary discrete time series (Phillips, 1998, 2001, Phillips and Park, 1998) and continuous-time diffusions (Bandi and Phillips, 2002, 2003). For a recurrent diffusion, $\overline{L}_X(T,x)$ diverges to infinity as $T \to \infty$ (*e.g.* it diverges with rates $\sqrt{T}$ and T for Brownian motions and stationary processes respectively). The local time defined by (2) will be used in what follows to characterize the convergence rates of various quantities just as the role of the sample size in traditional discrete-time stationary time series analysis. For a fixed time T, $\overline{L}_X(T,x)$ can be (strongly) consistently estimated by

$$\widehat{\overline{L}}_X(T,x,h) = \frac{\Delta}{h}\sum_{i=1}^{n} K(\frac{X_{i\Delta}-x}{h}). \tag{3}$$

Here $h = h_{n,T}$ is a bandwidth parameter depending on n and T (see Assumption A), and $K(\cdot)$ is a kernel function satisfying the following condition.

Assumption K. *The kernel function $K(\cdot)$ is a continuously differentiable, symmetric and bounded density function with a bounded support $(-1,1)$, such that $\int_{-1}^{1}|K'(u)|^2du < \infty$. Define $K_1 = \int_{-1}^{1} u^2K(u)du$, $K_2 = \int_{-1}^{1} K^2(u)du$.*

2.3 Nonparametric estimation of infinitesimal moment functions

Nonparametric estimation of the drift and diffusion functions is based on the following approximate conditional moment restrictions (see (4) and (5)). For a smooth function f, the conditional expected increment can be expressed as

$$\mathbb{E}(f(X_{t+\Delta}) - f(X_t)|X_t) = \mathcal{L}f(X_t)\Delta + \frac{1}{2}\mathcal{L}^2 f(X_t)\Delta^2 + O(\Delta^3),$$

where the infinitesimal generator of the process X_t, $\mathcal{L}$, is defined by $\mathcal{L}f(x) = f'(x)\mu(x) + \frac{1}{2}f''(x)\sigma^2(x)$. Letting $f(x) = x - X_t$, $(x - X_t)^2$, $(x - X_t)^4$ gives respectively

$$\mathbb{E}\left(\frac{X_{t+1}-X_t}{\Delta}|X_t\right) = \mu(X_t) + o(1), \tag{4}$$

$$\mathbb{E}\left(\frac{(X_{t+1}-X_t)^2}{\Delta}|X_t\right) = \sigma^2(X_t) + o(1), \tag{5}$$

and

$$\mathbb{E}\left(\frac{(X_{t+1}-X_t)^4}{\Delta^2}|X_t\right) = 3\sigma^4(X_t) + o(1). \tag{6}$$

as $\Delta \to 0$ (the condition (6) will be used in deriving the limit distribution of the empirical likelihood ratio for $\sigma^2(\cdot)$). For a state point $x \in \mathcal{D}$, the local polynomial estimators of the drift and diffusion functions are defined by $\widehat{\mu}(x,h) = \alpha_0$ and $\widehat{\sigma}^2(x,h) = \beta_0$, where $(\alpha_0, \cdots, \alpha_p)$ and $(\beta_0, \cdots, \beta_p)$ minimize the weighted sums of squares (resp.)

$$\sum_{i=1}^{n}\left(\frac{X_{(i+1)\Delta}-X_{i\Delta}}{\Delta} - \sum_{j=0}^{p}\alpha_j(X_{i\Delta}-x)^j\right)^2 K\left(\frac{X_{i\Delta}-x}{h}\right),$$

and

$$\sum_{i=1}^{n}\left(\frac{(X_{(i+1)\Delta}-X_{i\Delta})^2}{\Delta} - \sum_{j=0}^{p}\beta_j(X_{i\Delta}-x)^j\right)^2 K\left(\frac{X_{i\Delta}-x}{h}\right).$$

The Nadaraya-Watson (or local level) estimators and the local linear estimators are obtained by specifying $p = 0$ and $p = 1$, respectively, where the latter are the focus of the present paper and have the following explicit-form expressions

$$\widehat{\mu}(x,h) = \frac{\sum_{i=1}^{n} W_i(x,h)\frac{X_{(i+1)\Delta}-X_{i\Delta}}{\Delta}}{\sum_{i=1}^{n} W_i(x,h)}, \tag{7}$$

$$\widehat{\sigma}^2(x,h) = \frac{\sum_{i=1}^{n} W_i(x,h)\frac{(X_{(i+1)\Delta}-X_{i\Delta})^2}{\Delta}}{\sum_{i=1}^{n} W_i(x,h)}. \tag{8}$$

where $W_i(x,h) = K_h(X_{i\Delta}-x)S_{n,2} - (X_{i\Delta}-x)K_h(X_{i\Delta}-x)S_{n,1}$ with $S_{n,j} = \sum_{i=1}^{n}(X_{i\Delta}-x)^j K_h(X_{i\Delta}-x)$, $j = 1, 2$ and $K_h(\cdot) = K(\cdot/h)/h$.

Local linear smoothing is routinely used in nonparametric modeling. It has advantages

over the local level method such as appealing bias properties and minimax efficiency (Fan and Gijbels 1996) without inflating the variance of the underlying estimator like local higher order polynomial fitting. An undesirable property of the local linear estimator for the diffusion function is that it may produce unreasonable negative results. This is because the local linear weights are not always positive in finite samples especially when the bandwidth is large. An alternative volatility estimation method is proposed by Phillips and Xu (2006) and Xu (2006) to address this issue. The limit theory developed in Section 3 relies on the long span ($T \to \infty$) and in-fill ($\Delta \to 0$) asymptotics specified as follows.

Assumption A (Asymptotics). *Assume* $n \to \infty$, $T \to \infty$, $\Delta = \Delta_{n,T} \to 0$ *and* $h = h_{n,T} \to 0$ *such that*

$$\frac{\overline{L}_X(T,x)}{h}\sqrt{\Delta \log \frac{1}{\Delta}} = o_{a.s.}(1).$$

3 EL Point-wise Confidence Intervals

3.1 Confidence interval for the drift

This subsection describes a new approach to constructing the point-wise confidence interval for the drift function based on empirical likelihood (EL). Keep in mind that similar procedures can be applied to the diffusion function, which will be subsequently discussed in the next subsection.

We first motivate the EL approach in the moment restriction model. Suppose that the scalar parameter θ is identified by the moment condition

$$\mathbb{E}[g(Z,\theta)] = 0, \tag{9}$$

where g is a scalar real-valued function. Given *iid* observations $\{Z_1, \cdots, Z_n\}$, the method of moment estimates θ by substituting the population mean by the sample mean, *viz.* it solves the estimating equation

$$\frac{1}{n}\sum_{i=1}^{n} g(Z_i, \widehat{\theta}) = 0.$$

The empirical likelihood method looks at $\widehat{\theta}$ as the nonparametric maximum likelihood estimator (NPMLE). The concept was first introduced by Owen (1988, 1990) in the context of estimating the mean of an unknown distribution, *viz.* $g(Z,\theta) = Z - \theta$, and the NPMLE of the true value of θ is the sample average $n^{-1}\sum_{i=1}^{n} Z_i$. The idea is to treat $\{Z_i\}_{i=1}^{n}$ as if they are multinomial random variables with probability p_i assigned to each Z_i. Thus, the (nonparametric) log likelihood function is

$$\ell_{NP} = \sum_{i=1}^{n} \log(p_i),\ p_i \geq 0,\ \sum_{i=1}^{n} p_i = 1.$$

Obviously ℓ_{NP} is maximized at the empirical distribution of the sample, *viz.* $\{p_i = \frac{1}{n}, i = 1,..,n\}$, which, when plugged into (9), yields the NPMLE $\widehat{\theta}$. Inference about θ can be drawn by mimicking the traditional parametric likelihood ratio procedure. The data-driven (empirical) likelihood ratio for θ is defined as

$$R(\theta) = \frac{\max\limits_{(p_1,\cdots,p_n)} \left\{ \prod_{i=1}^{n} p_i | \sum_{i=1}^{n} p_i g(Z_i,\theta) = 0, p_i \geq 0, \sum_{i=1}^{n} p_i = 1 \right\}}{\prod_{i=1}^{n} n^{-1}}. \tag{10}$$

The numerator of (10) is called the empirical likelihood, which is the nonparametric likelihood subject to model restrictions. The denominator of (10) is the unconstrained maximized nonparametric likelihood, and we accept values of θ such that this ratio $R(\theta)$ is large (*i.e.*, close to unity). The EL-based inference procedure does not rely on parametric specification of an underlying distributional family and the resultant confidence intervals have the same order of coverage error as ones based on the correctly specified parametric likelihood. It also inherits certain advantages from likelihood-based methods, such as automatic determination of the shape of the confidence region and favorable power properties.

Empirical likelihood can be also used in curve fitting problems. Owen (2001, Chapter 5) used EL to construct point-wise confidence intervals around the Nadaraya-Watson kernel estimates. Chen and Qin (2000) considered EL combined with the local linear smoother and found that it compared favorably with the asymptotic normality based inference procedure. Other applications of Chen and Qin's technique can be found in Qin and Tsao (2005) and Chan

et al (2006). Noticing the close similarity between the traditional nonparametric regression and nonparametric continuous-time diffusion model estimation, we are able to construct the confidence interval for the drift function in this subsection (and the diffusion function in the next subsection) following Chen and Qin's method.

Let $g_{\mu i}(x,h,\theta) = W_i(x,h)\left(\frac{X_{(i+1)\Delta}-X_{i\Delta}}{\Delta} - \theta\right)$, where the $W_i(x,h)$'s are the local linear weights in the drift function estimator defined in (7). Thus, $\widehat{\mu}(x,h)$ satisfies the estimating equation

$$\frac{1}{n}\sum_{i=1}^{n} g_{\mu i}(x,h,\widehat{\mu}(x,h)) = 0,$$

which leads us to define the following empirical likelihood ratio statistic for a candidate value θ :

$$R_\mu(x,h,\theta) = \max_{(p_1,\cdots,p_n)} \left\{ \prod_{i=1}^{n} np_i | \sum_{i=1}^{n} p_i g_{\mu i}(x,h,\theta) = 0, p_i \geq 0, \sum_{i=1}^{n} p_i = 1 \right\}. \tag{11}$$

and $l_\mu(x,h,\theta) = -2\log R_\mu(x,h,\theta)$. Notice the (log) EL ratio for θ depends on the state point considered x and a smoothing bandwidth parameter h. The following theorem characterizes the asymptotic behavior of $l_\mu(x,h,\theta)$ when θ is local to the true value $\mu(x)$. We use " $\Rightarrow$ " to denote the weak convergence of a random sequence.

Theorem 1. *Suppose that Assumption E, R, K hold. Let the bandwidth parameter h be such that Assumption A holds. Furthermore, if*

$$h\overline{L}_X(T,x) \overset{a.s.}{\to} \infty \tag{12}$$

and

$$h^5\overline{L}_X(T,x) \overset{a.s.}{\to} 0, \tag{13}$$

as $n, T \to \infty$, then,

$$l_\mu(x,h,\mu(x)) \Rightarrow \chi^2(1), \tag{14}$$

and

$$l_\mu\left(x,h,\mu(x) + \frac{\tau(x)}{\sqrt{h\overline{L}_X(T,x)}}\right) \Rightarrow \chi^2\left(1, \frac{\tau^2(x)}{K_2\sigma^2(x)}\right), \tag{15}$$

as $n, T \to \infty$, *where* $\tau(x)$ *is fixed and* $\chi^2(p_1, p_2)$ *is the chi-squared distribution with degree of freedom* p_1 *and non-central parameter* p_2. *Then the* $100(1-\alpha)\%$ *EL confidence interval is defined as* $I^{EL}_{\mu,\alpha} = \{\theta : l_\mu(x, h, \theta) \leq \chi^2_{1-\alpha}(1)\}$, *where* $\chi^2_{1-\alpha}(1)$ *is the inverse cumulative distribution function (cdf) for the* $\chi^2(1)$ *distribution evaluated at* $1-\alpha$.

Remarks.

(1). In addition to Assumption A, two more conditions are imposed on the bandwidth parameter h. (12) is necessary for consistency of the local linear estimator for the drift. The undersmoothing condition (13) is assumed to eliminate the bias of $\widehat{\mu}(x, h)$. Of course, there are some alternatives to achieve this, including adjusting the confidence interval according to the explicitly estimated bias term. This alternative, however, "generally increases variance more than it reduces bias" (Sun and Loader, 1994), since it involves estimating the second derivative of $\mu(x)$ which usually cannot be estimated well even with large sample sizes. This is particularly true in continuous-time models since the drift function (and its derivatives) is difficult to estimate given the time span over typical data sets.

Thus, to construct an asymptotically valid confidence interval for the drift function, the bandwidth parameter h is chosen such that $h = O_{a.s.}((\overline{L}_X(T, x))^{-k})$, where $0.2 < k < 1$. When X_t is positive recurrent, which is implied by strict stationarity, h should satisfy $h = O(T^{-k})$, where $0.2 < k < 1$.

(2). The EL confidence interval is constructed by collecting all candidate values that can not be rejected by the test based on the statistic $l_\mu(x, h, \theta)$ at the specified level. It is made clear in (15) that the EL-based testing procedure can detect deviations from the true value of the drift function up to a quantity of order $O_p((h\overline{L}_X(T, x))^{-1/2})$. This fact shows that the resultant confidence interval shrinks only when the sample size increases by way of expanding the local time spent around the state point x. If we only obtain more observations over a fixed sampling interval without lengthening the time span, we can not get shorter confidence interval with the same confidence level.

(3). The quantity $l_\mu(x,h,\theta)$ has a closed-form representation and is computed easily. Using Lagrange multipliers, the constrained optimization problem in (11) is solved by

$$\widehat{p}_i = \frac{1}{n(1+\lambda_\mu(x,h,\theta)g_{\mu i}(x,h,\theta))}, \tag{16}$$

where $\lambda_\mu(x,h,\theta)$ satisfies

$$\sum_{i=1}^{n} \frac{g_{\mu i}(x,h,\theta)}{1+\lambda_\mu(x,h,\theta)g_{\mu i}(x,h,\theta)} = 0. \tag{17}$$

Thus, given a candidate parameter value θ, $l_\mu(x,h,\theta)$ can be evaluated by first solving (17) numerically (*e.g.* by Newton's method) and computing

$$l_\mu(x,h,\theta) = 2\sum_{i=1}^{n} \log\left(1+\lambda_\mu(x,h,\theta)g_{\mu i}(x,h,\theta)\right). \tag{18}$$

The EL confidence interval for $\mu(x)$ can be constructed using the following algorithm:

Algorithm (EL confidence interval for $\mu(x)$)

1. Select a step size s. Let $\theta_j = \widehat{\mu}(x,h) + js$, $j = 1, 2, \cdots$, and compute $l_\mu(x,h,\theta_j)$. Write j_1 as the first j such that $l_\mu(x,h,\theta_j) > \chi^2_{1-\alpha}(1)$. Let $\theta^U = \theta_{j_1-1}$.

2. Let $\theta_j = \widehat{\mu}(x,h) - js$, $j = 1, 2, \cdots$, and compute $l_\mu(x,h,\theta_j)$. Write j_2 as the first j such that $l_\mu(x,h,\theta_j) > \chi^2_{1-\alpha}(1)$. Let $\theta^L = \theta_{j_2-1}$.

3. Form the $100(1-\alpha)\%$ EL confidence interval for $\mu(x)$ as $I^{EL}_{\mu,\alpha} = (\theta^L, \theta^U)$.

Since j_1 and j_2 in the algorithm above are different in general, the EL confidence interval has a data-determined shape and is able to account for possible skewness in the sampling distributions.

(4). The proof of Theorem 1 is sketched as follows. When the log EL ratio $l_\mu(x,h,\theta)$ is evaluated at the true value $\mu(x)$, it can be written in the form of a self-normalized sum; we

will show that as $n, T \to \infty$,

$$\begin{aligned} l_\mu(x,h,\mu(x)) &= \left(\sum_{i=1}^n g_{\mu i}^2(x,h,\mu(x))\right)^{-1}\left(\sum_{i=1}^n g_{\mu i}(x,h,\mu(x))\right)^2 + o_p(1) \qquad (19)\\ &= A_\mu^{-1}(x,h,\mu(x))B_\mu^2(x,h,\mu(x)) + o_p(1), \end{aligned}$$

where

$$A_\mu(x,h,\theta) = \left(\sum_{i=1}^n W_i(x,h)\right)^{-2}\sum_{i=1}^n g_{\mu i}^2(x,h,\theta), \qquad (20)$$

$$B_\mu(x,h,\theta) = \left(\sum_{i=1}^n W_i(x,h)\right)^{-1}\sum_{i=1}^n g_{\mu i}(x,h,\theta), \qquad (21)$$

and

$$h\overline{L}_X(T,x)A_\mu(x,h,\mu(x)) \stackrel{a.s.}{\to} K_2\sigma^2(x), \qquad (22)$$

$$\sqrt{h\overline{L}_X(T,x)}B_\mu(x,h,\mu(x)) \Rightarrow \mathcal{N}(0, K_2\sigma^2(x)), \qquad (23)$$

where K_2 is a constant defined in Assumption K. Thus (14) follows from the fact that a squared standard normal random variable is χ^2- distributed with one degree of freedom. Note that $B_\mu(x,h,\mu(x)) = \widehat{\mu}(x,h) - \mu(x)$ and (23) follows essentially from the asymptotic normality of the local linear estimator $\widehat{\mu}(x,h)$, see, *e.g.*, Moloche (2001). The proof of Theorem 3.1 shows that EL implicitly chooses $A_\mu(x,h,\mu(x))$ as a deputy of the variance of $\widehat{\mu}(x,h)$. This natural variance estimator also utilizes the information contained in the true parameter value, which is not feasible in the construction of the conventional normal confidence interval. Chen and Qin (2000) suggested an EL-based symmetric normal confidence interval using the variance estimator $A_\mu(x,h,\widehat{\mu}(x,h))$.

3.2 Confidence interval for the diffusion

Now we turn to construction of the point-wise confidence interval for the diffusion function. Let $g_{\sigma i}(x,h,\theta) = W_i(x,h)\left(\frac{(X_{(i+1)\Delta}-X_{i\Delta})^2}{\Delta} - \theta\right)$. The local linear estimator of the diffusion

coefficient $\widehat{\sigma}^2(x,h)$ satisfies the estimating equation:

$$\frac{1}{n}\sum_{i=1}^{n} g_{\sigma i}(x,h,\widehat{\sigma}^2(x,h)) = 0.$$

Let

$$R_\sigma(x,h,\theta) = \max\left\{\prod_{i=1}^{n} np_i | \sum_{i=1}^{n} p_i g_{\sigma i}(x,h,\theta) = 0, p_i \geq 0, \sum_{i=1}^{n} p_i = 1\right\},$$

and $l_\sigma(x,h,\theta) = -2\log R_\sigma(x,h,\theta)$.

Theorem 2. *Suppose that Assumption E, R, K hold. Let the bandwidth parameter h be such that Assumption A holds. Furthermore, if*

$$\frac{h\overline{L}_X(T,x)}{\Delta} \overset{a.s.}{\to} \infty \tag{24}$$

$$\frac{h^5\overline{L}_X(T,x)}{\Delta} \overset{a.s.}{\to} 0, \tag{25}$$

as $n, T \to \infty$, then,

$$l_\sigma(x,h,\sigma^2(x)) \Rightarrow \chi^2(1),$$

and

$$l_\sigma\left(x,h,\sigma^2(x) + \frac{\tau(x)}{\sqrt{h\overline{L}_X(T,x)/\Delta}}\right) \Rightarrow \chi^2\left(1, \frac{\tau^2(x)}{2K_2\sigma^4(x)}\right),$$

as $n, T \to \infty$, where $\tau(x)$ is fixed and $\chi^2(p_1,p_2)$ is the chi-squared distribution with degree of freedom p_1 and non-central parameter p_2. Then the $100(1-\alpha)\%$ EL confidence interval is $I^{EL}_{\sigma,\alpha} = \{\theta > 0 : l_\sigma(x,h,\theta) \leq \chi^2_{1-\alpha}(1)\}$.

Remarks.

(1). As in the construction of the confidence interval for the drift function, the condition (24) is required for consistency of the local linear diffusion function estimator, and an undersmoothing bandwidth (see (25)) is chosen for bias elimination. Compared to the drift function case, a smaller bandwidth is used in constructing the confidence interval for the diffusion function. This reflects the fact that the nonparametric diffusion estimator converges with

a faster rate than the drift function estimator, thereby giving a shorter confidence interval for the diffusion function for the same sample of data. Here the bandwidth parameter h should satisfy $h = O_{a.s.}((\frac{\overline{L}_X(T,x)}{\Delta})^{-k})$, where $0.2 < k < 1$. When X_t is positive recurrent (implied by stationarity), h should satisfy $h = O(n^{-k})$, where $0.2 < k < 1$.

(2). The normal confidence interval for the diffusion function (described in detail in the next section) may cover the region with some negative values, causing inference difficulties in empirical applications. The common practice of circumventing this problem is to truncate the negative part of the interval. This is obviously unsatisfactory since it is no longer an interval estimate with the specified confidence level. This problem also occurs in the EL-based method, since the local linear estimator of the diffusion function may be negative, as illustrated in Section 2.3, and this value will be unavoidably included in the confidence interval because it has the largest nonparametric (empirical) likelihood. But it will be much less problematic since in our construction a small undersmoothing bandwidth is used which generally avoids negative local linear weights, thereby automatically confining the EL-based confidence interval for the diffusion function within the nonnegative region. This is essentially different from the normal confidence interval construction, in which case even with all positive local linear weights, this negativity problem still arises in finite samples.

(3). As in the drift case, the statistic $l_\sigma(x,h,\theta)$ can be computed explicitly. Using Lagrange multipliers,

$$l_\sigma(x,h,\theta) = \sum_{i=1}^{n} \log\Big(1 + \lambda_\sigma(x,h,\theta) g_{\sigma i}(x,h,\theta)\Big)$$

where $\lambda_\sigma(x,h,\theta)$ satisfies

$$\sum_{i=1}^{n} \frac{g_{\sigma i}(x,h,\theta)}{1 + \lambda_\sigma(x,h,\theta) g_{\sigma i}(x,h,\theta)} = 0.$$

It can be shown that as $n, T \to \infty$,

$$\begin{aligned} l_\sigma(x,h,\sigma^2(x)) &= \Big(\textstyle\sum_{i=1}^n g_{\sigma i}^2(x,h,\sigma^2(x))\Big)^{-1} \Big(\textstyle\sum_{i=1}^n g_{\sigma i}(x,h,\sigma^2(x))\Big)^2 + o_p(1) \qquad (26) \\ &= A_\sigma^{-1}(x,h,\sigma^2(x)) B_\sigma^2(x,h,\sigma^2(x)) + o_p(1), \end{aligned}$$

where

$$A_\sigma(x,h,\theta) = \left(\sum_{i=1}^n W_i(x,h)\right)^{-2} \sum_{i=1}^n g_{\sigma i}^2(x,h,\theta), \tag{27}$$

$$B_\sigma(x,h,\theta) = \left(\sum_{i=1}^n W_i(x,h)\right)^{-1} \sum_{i=1}^n g_{\sigma i}(x,h,\theta),$$

and

$$\frac{h\overline{L}_X(T,x)}{\Delta} A_\sigma(x,h,\sigma^2(x)) \overset{a.s.}{\to} 2K_2\sigma^4(x), \tag{28}$$

$$\sqrt{\frac{h\overline{L}_X(T,x)}{\Delta}} B_\sigma(x,h,\sigma^2(x)) \Rightarrow \mathcal{N}(0, 2K_2\sigma^4(x)). \tag{29}$$

Empirical likelihood implicitly uses $A_\sigma(x,h,\sigma^2(x))$ to estimate the variance of $\widehat{\sigma}^2(x,h)$.

4 Normal Confidence Intervals and Numerical Results

In this section we compare the finite sample performance of our EL based procedure in a numerical example with the conventional confidence interval construction method based on asymptotic normality, for both drift and diffusion functions.

Before describing our simulation designs, we outline the asymptotic normality-based method and focus on the drift function first. Suppose that Assumption E, R, K hold and the bandwidth parameter h satisfies Assumption A, (12) and (13). It can be shown that (23) holds, rewritten as

$$\sqrt{h\overline{L}_X(T,x)}(\widehat{\mu}(x,h) - \mu(x)) \Rightarrow \mathcal{N}(0, K_2\sigma^2(x)),$$

as $n, T \to \infty$. The quantities $\overline{L}_X(T,x)$ and $\sigma^2(x)$ are estimated by $\widehat{\overline{L}}_X(T,x,h_{lt})$ and $\widehat{\sigma}^2(x,h_{diff})$ defined in (3) and (8) respectively. Here the bandwidth parameters h_{lt} and h_{diff} satisfy Assumption A such that $\widehat{\overline{L}}_X(T,x,h_{lt})$ and $\widehat{\sigma}^2(x,h_{diff})$ consistently estimate $\overline{L}_X(T,x)$ and $\sigma^2(x)$ respectively. Hence the normal $100(1-\alpha)\%$ level confidence interval for $\mu(x)$ is given by

$$I_{\mu,\alpha}^{ASY} = \left[\widehat{\mu}(x,h) - z_{1-\alpha/2}\left(\widehat{var}(\widehat{\mu}(x,h))\right)^{1/2},\ \widehat{\mu}(x,h) + z_{1-\alpha/2}\left(\widehat{var}(\widehat{\mu}(x,h))\right)^{1/2}\right],$$

where $\widehat{var}(\widehat{\mu}(x,h)) = \left(h\widehat{\overline{L}}_X(T,x,h_{lt})\right)^{-1} K_2\widehat{\sigma}^2(x,h_{diff})$, and $z_{1-\alpha/2}$ is the inverse cdf for the

standard normal distribution evaluated at $1-\alpha/2$. Note that here we use an undersmoothing bandwidth h in the local linear estimator, but h_{lt} and h_{diff} used in estimating the local time and the diffusion function respectively may be different from h. Usually h_{lt} and h_{diff} are chosen in practice as "optimal" such that the squared bias and the variance of the associated estimator are of the same order.

In a similar way, the normal $100(1-\alpha)\%$ level confidence interval for $\sigma^2(x)$ is given by

$$I_{\sigma,\alpha}^{ASY} = \left[\widehat{\sigma}^2(x,h) - z_{1-\alpha/2}\left(\widehat{var}(\widehat{\sigma}^2(x,h))\right)^{1/2},\ \widehat{\sigma}^2(x,h) + z_{1-\alpha/2}\left(\widehat{var}(\widehat{\sigma}^2(x,h))\right)^{1/2}\right] \cap (0,+\infty),$$

where $\widehat{var}(\widehat{\sigma}^2(x,h)) = 2\left(\frac{h\widehat{\overline{L}}_X(T,x,h_{lt})}{\Delta}\right)^{-1} K_2\widehat{\sigma}^4(x,h_{diff})$. As mentioned in Section 3.2, truncation will reduce, often substantially, the confidence level of the normal interval for the diffusion function. In general, for both drift and diffusion functions, the normal confidence interval constructed above relies on the estimation of separate parts of the asymptotic variance, and is usually inaccurate and sensitive to bandwidth choices. Moreover, it is artificially centered at the corresponding local linear estimate, failing to account for the potential asymmetry in the distribution of the estimate in small samples.

In our Monte Carlo experiments, the data generating process is the CIR or square-root model (Cox, Ingersoll and Ross, 1985):

$$dX_t = \kappa(\theta - X_t)dt + \sigma\sqrt{X_t}dB_t,$$

where the parameter values $\kappa = 0.214592$, $\theta = 0.085711$, $\sigma = 0.07830$ are taken from Chapman and Pearson (2000). We generate 30 ($T = 30$) and 60 ($T = 60$) years of daily ($\Delta = 1/252$) data by generating $X_{t+\Delta}$ from the transition density $(2c)^{-1}\chi^2(2u, 2g+2)$ given X_t, where $c = (\sigma^2(1-e^{-\kappa\Delta}))^{-1}2\kappa$, $u = cX_te^{\kappa\Delta}$ and $g = 2\kappa\theta\sigma^{-2} - 1 \geq 0$. So the sizes of the samples we use are $n = 7500, 15000$. To check the robustness of the proposed inference procedures to the stationarity assumption, we set the initial value $X_0 = 0.067$ instead of taking it from the invariant distribution. The first 500 observations are discarded to eliminate the effects of initiation. The random number generator of non-central chi-squared distribution we use is the `ncx2rnd` function of Matlab 7.0.

To estimate the model based on the data generated, we use the Epanechnikov kernel function:

$$K(u) = \begin{cases} 0.75(1-u^2), & -1 < u < 1 \\ 0, & \text{otherwise} \end{cases}.$$

For the drift function, we use 25 equal-spaced bandwidths between $0.03T^{-1/3}$ and $0.15T^{-1/3}$, and for the diffusion function, we use 24 equal-spaced bandwidths between $0.11n^{-1/3}$ and $0.34n^{-1/3}$. Let $\widehat{s}$ be the standard deviation of the sample $\{X_t, t = 1, \cdots, n\}$. The bandwidths h_{lt} and h_{diff} used in normal confidence intervals are chosen as $h_{lt} = h_{diff} = \widehat{s}n^{-1/5}$, which are employed by Chapman and Pearson (2000) and shown as optimal bandwidths. The three design points we consider are $x = 0.04, 0.08, 0.14$, which are usually within the empirical range of most U.S. interest rate data sets. All the results reported in this section are for confidence intervals conducted at the 95% nominal confidence level (Figure 2-9), and are computed over 5000 replications.

Before moving to the confidence interval construction, we first compare the null distribution of the two test statistics based on which confidence intervals are constructed. Figure 1 plots density functions of the log EL ratios (solid line) and the squared $t-$statistics (dashed line marked with " + ") evaluated at the true values of the drift (Figure 1a and 1b) and diffusion functions (Figure 1c and 1d), based on 5000 samples of 30 years ($T = 30$) of daily data ($1/\Delta = 252$). The state point considered is $x = 0.04$. Two different fixed bandwidths are used for the drift ($h = 0.02, 0.04$) and diffusion functions ($h = 0.01, 0.02$). Density functions of these two statistics based on samples generated are compared with their common limiting $\chi^2(1)$ density function (dashed line). These figures show that in general, for both drift and diffusion functions and both two bandwidths considered, the finite sample density functions of the squared $t-$statistics have much heavier tails than the $\chi^2(1)$ density, especially when the bandwidths are large. However, the finite sample distribution of the log EL ratio statistic is reasonably well approximated by its limiting $\chi^2(1)$ distribution.

Figure 1 about here

In Figure 2 and 3 we report the empirical coverage probabilities and lengths (after coverage level calibration) of point-wise confidence intervals for the drift function against various bandwidths using two procedures based on EL and asymptotic normality. The corresponding results for the diffusion function are plotted in Figure 4 and 5. All these four figures share the same structure. The left and right columns of the graphs report results with sample sizes 7500 and 15000, respectively, and the three rows corresponds to the three states that the process visits: 0.04, 0.08 and 0.14. The solid and dotted lines represent reported quantities for EL and normal confidence intervals respectively.

Figure 2-5 about here

Several observations are in order. Consider confidence intervals for the drift function first. Figure 2 shows that the EL based inference procedure is very accurate; the empirical coverage rates are very close to the nominal confidence level 95%, for all bandwidths and design points considered. For example, for $x = 0.04$, the coverage probabilities of the EL intervals over 25 bandwidths are between 93.4% and 94.9%, with an average 94.2% when $T = 30$, and the range becomes between 94.5% and 95.9% and the average is 95.2% when $T = 60$. Conversely, normal intervals are seriously under-covered; for $x = 0.04$, the coverage rates of the normal intervals over 25 bandwidths are between 79.9% and 85.1%, with an average 82.5% when $T = 30$, and the range is between 81.4% and 90.8% and the average is 85.1% when $T = 60$. Figure 3 shows that the adjusted length (after confidence level calibration) of the EL interval is uniformly shorter than that of the normal interval for every bandwidth considered, although the difference between them becomes less as the time span increases.

For the diffusion function, Figure 4 shows that the EL confidence intervals are also accurate, just as those for the drift, for all bandwidths and both time spans under consideration. The normal intervals are generally subject to under-coverage, but less pronounced than those for the drift function. For most cases, the performance of EL intervals clearly dominates that for normal intervals, although they appear to be much closer than those for the drift function. For instance, for $x = 0.08$, the averages of the coverage probabilities for the EL intervals and normal intervals over 24 bandwidths considered are 94.0% and 95.1% respectively for $T = 30$, and the

averages become 94.9% and 95.0% respectively when $T = 60$. Figure 5 gives the adjusted lengths (after confidence level calibrations) for the EL and normal confidence intervals for the diffusion function. EL intervals are generally shorter than normal intervals, but again, the differences are less than the cases for the drift function.

We repeat the experiments above with more sparsely sampled data - weekly data ($\Delta = 1/52$) over 40 and 80 years ($T = 40, 80$). The corresponding results are reported in Figure 6-9. Advantages of the EL-based confidence intervals are clear, and strikingly their empirical coverage rates are still very close to the nominal confidence level 95%. Compared to the cases using daily data, the (EL and normal, both) confidence intervals here for the drift function are more concentrated because we use data with longer span although with fewer observations.

Figure 6-9 about here

In summary, we compared two different confidence interval construction procedures using data simulated from a square-root diffusion process. Parameter values are taken from the empirical literature of fitting real interest rate data. Other nonlinear diffusion models such as those in Aït-Sahalia (1996b) were also considered in our experiments, although not reported here, and they yield the results similar to those obtained above. Normal confidence intervals based on local linear fitting usually produce inaccurate coverage rates, especially for the drift function. Multiple bandwidth choices in the underlying local linear estimator and its variance estimator complicate the inference procedure. It is hard to select an appropriate variance estimator that can reflect the variability of the underlying estimate. The EL procedure circumvents this practical problem by studentizing internally, avoiding variance estimation. Our simulation studies show that the EL-based confidence interval performs very well and has clear advantages over the normal interval.

5 Conclusions

In this paper, we describe a new inference procedure for use in modeling nonparametric and potentially nonstationary continuous-time diffusions. The method is computationally easy to

implement and has other properties that make it appealing in practical applications. Simulation studies demonstrate many advantages over the conventional inference method based on asymptotic normality, including automatic data-determined shape characteristics, more accurate coverage probabilities and shorter confidence intervals.

6 Appendix: Proofs

In this section, we prove the main results in this paper, Theorems 1 and 2. Let C be a generic positive constant. The integral is always taken over the real line unless specified otherwise. For simplicity, in what follows, we omit notations of explicit dependence on the state point x and the smoothing bandwidth h, *e.g.*, we write $A_\mu(x,h,\theta)$, $g_{\mu i}(x,h,\theta)$, $l_\mu(x,h,\theta)$, $\lambda_\mu(x,h,\theta)$ and $W_i(x,h)$ simply as $A_\mu(\theta)$, $g_{\mu i}(\theta)$, $l_\mu(\theta)$, $\lambda_\mu(\theta)$ and $W_i(x)$ respectively.

Lemma A.1. Let $A_\mu(\theta)$ be defined in (20). Then $A_\mu(\theta) \overset{a.s.}{\to} K_2\sigma^2(x)$ for any fixed θ.

Proof. Note that $A_\mu(\theta)$ can be decomposed into three parts:

$$\begin{aligned}
A_\mu(\theta) &= \frac{\overline{L}_X(T,x)h\sum_{i=1}^n g_{\mu i}^2(\theta)}{\left(\sum_{i=1}^n W_i(x)\right)^2} \\
&= \frac{\frac{\overline{L}_X(T,x)h}{\Delta}\sum_{i=1}^n W_i^2(x)\frac{(X_{(i+1)\Delta}-X_{i\Delta})^2}{\Delta}}{\left(\sum_{i=1}^n W_i(x)\right)^2} - 2\Delta\frac{\theta\frac{\overline{L}_X(T,x)h}{\Delta}\sum_{i=1}^n W_i^2(x)\frac{X_{(i+1)\Delta}-X_{i\Delta}}{\Delta}}{\left(\sum_{i=1}^n W_i(x)\right)^2} \\
&\quad +\Delta\frac{\theta^2\frac{\overline{L}_X(T,x)h}{\Delta}\sum_{i=1}^n W_i^2(x)}{\left(\sum_{i=1}^n W_i(x)\right)^2} \\
&:= A_{\mu 1} + A_{\mu 2} + A_{\mu 3}. \qquad (30)
\end{aligned}$$

Look at $A_{\mu 1}$ first. By the definition of the local linear weights $W_i(x)$,

$$\begin{aligned} A_{\mu 1} &= \frac{\frac{\overline{L}_X(T,x)}{\Delta h} S_{n,2}^2 \sum_{i=1}^n K^2(\frac{X_{i\Delta}-x}{h}) \frac{(X_{(i+1)\Delta}-X_{i\Delta})^2}{\Delta}}{\left(\sum_{i=1}^n W_i(x)\right)^2} \\ &\quad -2\frac{\frac{\overline{L}_X(T,x)}{\Delta} S_{n,2} S_{n,1} \sum_{i=1}^n \frac{X_i-x}{h} K^2(\frac{X_{i\Delta}-x}{h}) \frac{(X_{(i+1)\Delta}-X_{i\Delta})^2}{\Delta}}{\left(\sum_{i=1}^n W_i(x)\right)^2} \\ &\quad +\frac{\frac{\overline{L}_X(T,x)h}{\Delta} S_{n,1}^2 \sum_{i=1}^n \frac{(X_i-x)^2}{h^2} K^2(\frac{X_{i\Delta}-x}{h}) \frac{(X_{(i+1)\Delta}-X_{i\Delta})^2}{\Delta}}{\left(\sum_{i=1}^n W_i(x)\right)^2} \\ &:= A_{\mu 11} + A_{\mu 12} + A_{\mu 13}. \end{aligned} \tag{31}$$

Before examining these three terms, we first establish the following result:

$$\frac{\sum_{i=1}^n W_i(x)}{\left(\sum_{i=1}^n K(\frac{X_{i\Delta}-x}{h})\right)^2} = \frac{S_{n,2}/h}{\sum_{i=1}^n K(\frac{X_{i\Delta}-x}{h})} - \left(\frac{S_{n,1}}{\sum_{i=1}^n K(\frac{X_{i\Delta}-x}{h})}\right)^2 \overset{a.s.}{\to} K_1. \tag{32}$$

Following the arguments in BP (p. 266-267),

$$\begin{aligned} \frac{S_{n,2}/h}{\sum_{i=1}^n K(\frac{X_{i\Delta}-x}{h})} &= \frac{\sum_{i=1}^n \frac{(X_{i\Delta}-x)^2}{h^2} K(\frac{X_{i\Delta}-x}{h})}{\sum_{i=1}^n K(\frac{X_{i\Delta}-x}{h})} = \frac{\int_0^T \frac{(X_s-x)^2}{h^3} K(\frac{X_s-x}{h}) ds + o_{a.s.}(1)}{\int_0^T \frac{1}{h} K(\frac{X_s-x}{h}) ds + o_{a.s.}(1)} \\ &= \frac{\int_0^T \frac{(y-x)^2}{h^3} K(\frac{y-x}{h}) s(y) dy + o_{a.s.}(1)}{\int_0^T \frac{1}{h} K(\frac{y-x}{h}) s(y) dy + o_{a.s.}(1)} \overset{a.s.}{\to} \frac{\int u^2 K(u) du \cdot s(x)}{\int K(u) du \cdot s(x)} = K_1, \end{aligned} \tag{33}$$

by Lemma 6 in BP and dominated convergence as $n, T \to \infty$. Similarly, it can be shown that

$$\frac{S_{n,1}}{\sum_{i=1}^n K(\frac{X_{i\Delta}-x}{h})} = h \cdot \frac{K_1 s'(x)}{s(x)} + O_{a.s.}(h^2) \overset{a.s.}{\to} 0, \tag{34}$$

by the symmetry of the kernel function. So (32) holds. Now look at the term $A_{\mu 11}$ in (31). By

(32),

$$
\begin{aligned}
A_{\mu 11} &= \frac{1}{K_1^2}\frac{\overline{L}_X(T,x)}{\frac{\Delta}{h}\sum_{i=1}^n K(\frac{X_{i\Delta}-x}{h})}\left(\frac{S_{n,2}/h}{\sum_{i=1}^n K(\frac{X_{i\Delta}-x}{h})}\right)^2 \frac{\sum_{i=1}^n K^2(\frac{X_{i\Delta}-x}{h})\frac{(X_{(i+1)\Delta}-X_{i\Delta})^2}{\Delta}}{\sum_{i=1}^n K(\frac{X_{i\Delta}-x}{h})} \\
&\overset{a.s.}{\to} \frac{1}{K_1^2}\cdot 1\cdot K_1^2\cdot K_2\sigma^2(x) \\
&= K_2\sigma^2(x), \qquad (35)
\end{aligned}
$$

where we use

$$
\frac{\overline{L}_X(T,x)}{\frac{\Delta}{h}\sum_{i=1}^n K(\frac{X_{i\Delta}-x}{h})} \overset{a.s.}{\to} 1 \qquad (36)
$$

and

$$
\frac{\sum_{i=1}^n K^2(\frac{X_{i\Delta}-x}{h})\frac{(X_{(i+1)\Delta}-X_{i\Delta})^2}{\Delta}}{\sum_{i=1}^n K(\frac{X_{i\Delta}-x}{h})} \overset{a.s.}{\to} K_2\sigma^2(x),
$$

the latter of which can be shown following the proof of Theorem 2 in BP. Similarly, we have

$$
\begin{aligned}
A_{\mu 12} &= -2\frac{\overline{L}_X(T,x)}{\frac{\Delta}{h}\sum_{i=1}^n K(\frac{X_{i\Delta}-x}{h})}\frac{S_{n,2}/h}{\sum_{i=1}^n K(\frac{X_{i\Delta}-x}{h})}\frac{S_{n,1}}{\sum_{i=1}^n K(\frac{X_{i\Delta}-x}{h})}\cdot \\
&\quad \cdot\frac{\sum_{i=1}^n \frac{X_i-x}{h}K^2(\frac{X_{i\Delta}-x}{h})\frac{(X_{(i+1)\Delta}-X_{i\Delta})^2}{\Delta}}{\sum_{i=1}^n K(\frac{X_{i\Delta}-x}{h})} \\
&\overset{a.s.}{\to} -2\cdot 1\cdot K_1\cdot 0\cdot\sigma^2(x)\int uK^2(u)du = 0 \qquad (37)
\end{aligned}
$$

and

$$
\begin{aligned}
A_{\mu 13} &= \frac{\overline{L}_X(T,x)}{\frac{\Delta}{h}\sum_{i=1}^n K(\frac{X_{i\Delta}-x}{h})}\left(\frac{S_{n,1}}{\sum_{i=1}^n K(\frac{X_{i\Delta}-x}{h})}\right)^2 \frac{\sum_{i=1}^n \frac{(X_i-x)^2}{h^2}K^2(\frac{X_{i\Delta}-x}{h})\frac{(X_{(i+1)\Delta}-X_{i\Delta})^2}{\Delta}}{\sum_{i=1}^n K(\frac{X_{i\Delta}-x}{h})} \\
&\overset{a.s.}{\to} 1\cdot 0\cdot\sigma^2(x)\int u^2K^2(u)du \\
&= 0. \qquad (38)
\end{aligned}
$$

So $A_{\mu 1} \overset{a.s.}{\to} K_2\sigma^2(x)$. In a similar way it can be shown that $A_{\mu 2}, A_{\mu 3} = O_{a.s.}(\Delta) \overset{a.s.}{\to} 0$. Combining (35), (37), (38) and (30), Lemma A.1 holds.

Lemma A.2. Let $A_\sigma(\theta)$ be defined in (27). Then $A_\sigma(\theta) \overset{a.s.}{\to} 2K_2\sigma^4(x)$ for any fixed θ.

Proof. Write

$$
\begin{aligned}
A_\sigma(\theta) &= \frac{\frac{\overline{L}_X(T,x)h}{\Delta}\sum_{i=1}^n g_{\sigma i}^2(\theta)}{\left(\sum_{i=1}^n W_i(x)\right)^2} \\
&= \frac{\frac{\overline{L}_X(T,x)h}{\Delta}\sum_{i=1}^n W_i^2(x)\frac{(X_{(i+1)\Delta}-X_{i\Delta})^4}{\Delta^2}}{\left(\sum_{i=1}^n W_i(x)\right)^2} - 2\frac{\sigma^2(x)\frac{\overline{L}_X(T,x)h}{\Delta}\sum_{i=1}^n W_i^2(x)\frac{(X_{(i+1)\Delta}-X_{i\Delta})^2}{\Delta}}{\left(\sum_{i=1}^n W_i(x)\right)^2} \\
&\quad + \frac{\sigma^4(x)\frac{\overline{L}_X(T,x)h}{\Delta}\sum_{i=1}^n W_i^2(x)}{\left(\sum_{i=1}^n W_i(x)\right)^2} \\
&:= A_{\sigma 1} + A_{\sigma 2} + A_{\sigma 3}. \qquad (39)
\end{aligned}
$$

Noticing the moment condition in (6), it can be shown that $A_{\sigma 1} \overset{a.s.}{\to} 3K_2\sigma^4(x)$, $A_{\sigma 2} \overset{a.s.}{\to} -2K_2\sigma^4(x)$ and $A_{\sigma 3} \overset{a.s.}{\to} K_2\sigma^4(x)$, following similar steps to those in the proof of Lemma A.1. Thus we have $A_\sigma(\theta) \overset{a.s.}{\to} 2K_2\sigma^4(x)$ for any fixed θ.

Proof of Theorem 1. Recall that $g_{\mu i}(\theta) = W_i(x)\left(\frac{X_{(i+1)\Delta}-X_{i\Delta}}{\Delta} - \theta\right)$.

(a). In this part, we will prove that

$$
l_\mu(\theta) = \left(\sum_{i=1}^n g_{\mu i}^2(\theta)\right)^{-1}\left(\sum_{i=1}^n g_{\mu i}(\theta)\right)^2 + o_p(1) \qquad (40)
$$

holds for $\theta = \mu(x) + \frac{\tau(x)}{\sqrt{h\overline{L}_X(T,x)}}$ with any fixed $\tau(x)$, which includes (19) as a special case. To show (40), we need the following three results:

$$
\sup_{1\le i\le n} |g_{\mu i}(\theta)| = O_{a.s.}\left(\frac{\overline{L}_X(T,x)h}{\Delta^2}\sqrt{\Delta \log\frac{1}{\Delta}}\right), \qquad (41)
$$

$$
\lambda_\mu(\theta) = (\sum_{i=1}^n g_{\mu i}^2(\theta))^{-1}(\sum_{i=1}^n g_{\mu i}(\theta)) + o_p(1), \qquad (42)
$$

and

$$
|\lambda_\mu(\theta)| = O_p\left(\frac{\Delta^2}{(\overline{L}_X(T,x)h)^{1.5}}\right) = o_p(1), \qquad (43)
$$

with θ local to $\mu(x)$.

Provided the three results above are true, it follows from (41) and (43) that

$$\sup_{1\le i\le n} |\lambda_\mu(\theta) g_{\mu i}(\theta)| = O_p\left(\frac{1}{\sqrt{\overline{L}_X(T,x)h}}\sqrt{\Delta \log \frac{1}{\Delta}}\right) = o_p(1). \tag{44}$$

By (36) and (32),

$$\sum_{i=1}^n K(\frac{X_{i\Delta}-x}{h}) \overset{a.s.}{\sim} \frac{\overline{L}_X(T,x)h}{\Delta}, \tag{45}$$

$$\sum_{i=1}^n W_i(x) \overset{a.s.}{\sim} \frac{\overline{L}_X(T,x)h}{\Delta}, \tag{46}$$

where for two quantities A and B, $A \overset{a.s.}{\sim} B$ (or $A \overset{p}{\sim} B$) means that the ratio A/B goes to a finite nonzero number with probability one (or in probability). By Taylor expansion of the logarithmic function,

$$\log(1+\lambda_\mu(\theta)g_{\mu i}(\theta)) = \lambda_\mu(\theta)g_{\mu i}(\theta) - \frac{1}{2}\lambda_\mu^2(\theta)g_{\mu i}^2(\theta) + \xi_{\mu i}(\theta), \tag{47}$$

where the term $\xi_{\mu i}(\theta)$ is such that for some finite number $M>0$, $P(|\xi_{\mu i}(\theta)| \le M|\lambda_\mu(\theta)g_{\mu i}(\theta)|^3) \to 1$, $\forall 1\le i\le n$. In fact, by Lemma A1 and (46),

$$\sum_{i=1}^n g_{\mu i}^2(\theta) \overset{p}{\sim} \frac{(\overline{L}_X(T,x)h)^3}{\Delta^4}, \tag{48}$$

and thus $\sum_{i=1}^n |\lambda_\mu^3(\theta)g_{\mu i}^3(\theta)| \le \sup_{1\le i\le n}|\lambda_\mu(\theta)g_{\mu i}(\theta)|\cdot\sum_{i=1}^n \lambda_\mu^2(\theta)g_{\mu i}^2(\theta) = o_p(1)\cdot O_p\left(\frac{\Delta^4}{(\overline{L}_X(T,x)h)^3}\right)\cdot O_p\left(\frac{(\overline{L}_X(T,x)h)^3}{\Delta^4}\right) = o_p(1)$, by (44), (43) and (48). So $\sum_{i=1}^n \xi_{\mu i}(\theta) = o_p(1)$. Hence by (18), (47) and (42),

$$\begin{aligned}
-2\log R_\mu(\theta) &= 2\sum_{i=1}^n \log\Big(1+\lambda_\mu(\theta)g_{\mu i}(\theta)\Big) \\
&= 2\lambda_\mu(\theta)\sum_{i=1}^n g_{\mu i}(\theta) - \lambda_\mu^2(\theta)\sum_{i=1}^n g_{\mu i}^2(\theta) + 2\sum_{i=1}^n \xi_{\mu i}(\theta) \\
&= \left(\textstyle\sum_{i=1}^n g_{\mu i}^2(\theta)\right)^{-1}\left(\textstyle\sum_{i=1}^n g_{\mu i}(\theta)\right)^2 + o_p(1)
\end{aligned}$$

Thus (40) holds.

We conclude this part by showing (41) - (43). Look at (41) first. By (33), (34) and (45), $S_{n,1}, S_{n,2} = O_{a.s.}(\overline{L}_X(T,x)h^2/\Delta)$. Hence

$$\begin{aligned}
\sup_{1\le i\le n}|g_{\mu i}(\theta)| &= \sup_{1\le i\le n}|W_i(x)|\cdot\sup_{1\le i\le n}\left|\tfrac{X_{(i+1)\Delta}-X_{i\Delta}}{\Delta}-\theta\right| \\
&= \left(|S_{n,2}\sup_{1\le i\le n}K_h(X_{i\Delta}-x)|+|S_{n,1}\sup_{1\le i\le n}(X_i-x)K_h(X_{i\Delta}-x)|\right)\cdot \\
&\quad\cdot\sup_{1\le i\le n}\left|\tfrac{X_{(i+1)\Delta}-X_{i\Delta}}{\Delta}-\theta\right| \\
&= (O_{a.s.}(\frac{\overline{L}_X(T,x)h}{\Delta})+O_{a.s.}(\frac{\overline{L}_X(T,x)h^2}{\Delta}))\cdot O_{a.s.}(\frac{\kappa_{n,T}}{\Delta}) \\
&= O_{a.s.}\left(\tfrac{\overline{L}_X(T,x)h}{\Delta^2}\sqrt{\Delta\log\tfrac{1}{\Delta}}\right),
\end{aligned}$$

where $\kappa_{n,T} = \sup_{1\le i\le n}\left|X_{(i+1)\Delta}-X_{i\Delta}\right| = O_{a.s.}(\sqrt{\Delta\log\frac{1}{\Delta}})$ by the Lévy modulus of continuity of diffusions (see BP and Karatzas and Shreve, 1991).

Now we show (43). Let $g_{\mu j}(\theta) = \sum_{i=1}^n g_{\mu i}^j(\theta)$. By (16) and $\sum_{i=1}^n \widehat{p}_{\mu i} = 1$,

$$\sum_{i=1}^n \frac{1}{n(1+\lambda_\mu(\theta)g_{\mu i}(\theta))} = 1,$$

or

$$\frac{1}{n}\sum_{i=1}^n \frac{-g_{\mu i}(\theta)}{1+\lambda g_{\mu i}(\theta)} = 0, \tag{49}$$

and therefore (noting that $1+\lambda_\mu(\theta)g_{\mu i}(\theta) = (n\widehat{p}_{\mu i})^{-1} \ge 0$)

$$\begin{aligned}
0 &= \left|\tfrac{1}{n}\textstyle\sum_{i=1}^n\left(\tfrac{\lambda_\mu(\theta)g_{\mu i}^2(\theta)}{1+\lambda_\mu(\theta)g_{\mu i}(\theta)}-g_{\mu i}(\theta)\right)\right| \\
&\ge \frac{1}{n}\sum_{i=1}^n\frac{|\lambda_\mu(\theta)|g_{\mu i}^2(\theta)}{|1+\lambda_\mu(\theta)g_{\mu i}(\theta)|}-\left|\tfrac{1}{n}\textstyle\sum_{i=1}^n g_{\mu i}(\theta)\right| \\
&\ge \frac{|\lambda_\mu(\theta)|g_{\mu 2}(\theta)}{1+\sup_{1\le i\le n}|g_{\mu i}(\theta)|\cdot|\lambda_\mu(\theta)|}-|g_{\mu 1}(\theta)| \\
&= \frac{(g_{\mu 2}(\theta)-\sup_{1\le i\le n}|g_{\mu i}(\theta)|\cdot|g_{\mu 1}(\theta)|)|\lambda_\mu(\theta)|-|g_{\mu 1}(\theta)|}{1+C|\lambda_\mu(\theta)|},
\end{aligned}$$

since for any i, $|1+\lambda_\mu(\theta)g_i(\theta)| \le 1+|\lambda_\mu(\theta)|C$. Thus, we have

$$(g_{\mu 2}(\theta) - \sup_{1\le i\le n} |g_{\mu i}(\theta)|\cdot|g_{\mu 1}(\theta)|)|\lambda_\mu(\theta)| \le |g_{\mu 1}(\theta)|. \tag{50}$$

By (53) below, $\overline{L}_X(T,x)h \cdot \frac{g_{\mu 1}(\theta)}{\sum_{i=1}^n W_i(x)} = O_p(1)$ and so

$$|g_{\mu 1}(\theta)| = O_p(\frac{(\overline{L}_X(T,x)h)^{1.5}}{\Delta^2}) \tag{51}$$

in view of (46). By (41),

$$\sup_{1\le i\le n} |g_{\mu i}(\theta)|\cdot|g_{\mu 1}(\theta)| = O_p(\frac{(\overline{L}_X(T,x)h)^{2.5}}{\Delta^4}\sqrt{\Delta\log\frac{1}{\Delta}}). \tag{52}$$

Hence combining (48), (50), (51) and (52), (43) holds.

It remains to show (42). By (17),

$$\begin{aligned} 0 &= \frac{1}{n}\sum_{i=1}^n \frac{g_{\mu i}(\theta)}{1+\lambda_\mu(\theta)g_{\mu i}(\theta)} = \frac{1}{n}\sum_{i=1}^n g_{\mu i}(\theta)\left(1-\lambda_\mu(\theta)g_{\mu i}(\theta)+\lambda_\mu(\theta)^2 g_{\mu i}(\theta)^2\frac{1}{1+\lambda_\mu(\theta)g_{\mu i}(\theta)}\right) \\ &= \frac{\Delta^2}{n}g_{\mu 1}(\theta) - \frac{\Delta^2}{n}\lambda_\mu(\theta)g_{\mu 2}(\theta) + \frac{\Delta^2}{n}\varrho, \end{aligned}$$

where $\varrho = \sum_{i=1}^n \frac{\lambda_\mu(\theta)^2 g_{\mu i}(\theta)^3}{1+\lambda_\mu(\theta)g_{\mu i}(\theta)}$. By (41) and (43),

$$\frac{\Delta^2}{n}|\varrho| \le |\lambda_\mu(\theta)|^2\cdot\sup_{1\le i\le n}|g_{\mu i}(\theta)|^3\cdot O_p(1) = o_p(1)$$

Thus

$$\lambda_\mu(\theta) = \frac{g_{\mu 1}(\theta)}{g_{\mu 2}(\theta)} + o_p(1).$$

Hence (41) - (43) are true and (40) is proved.

(b) It follows immediately from the limit theory for the *undersmoothing* local linear estimate $\widehat{\mu}(x)$ (see, *e.g.* Moloche, 2001) that (23) holds and

$$B_\mu\left(\mu(x) + \frac{\tau(x)}{\sqrt{h\overline{L}_X(T,x)}}\right) \Rightarrow \mathcal{N}(\tau(x), K_2\sigma^2(x)). \tag{53}$$

Then (14) and (15) follows from (40), (53) and Lemma A.1.

Proof of Theorem 2. The proof follows that of Theorem 1 using Lemma A.2. and is omitted here.

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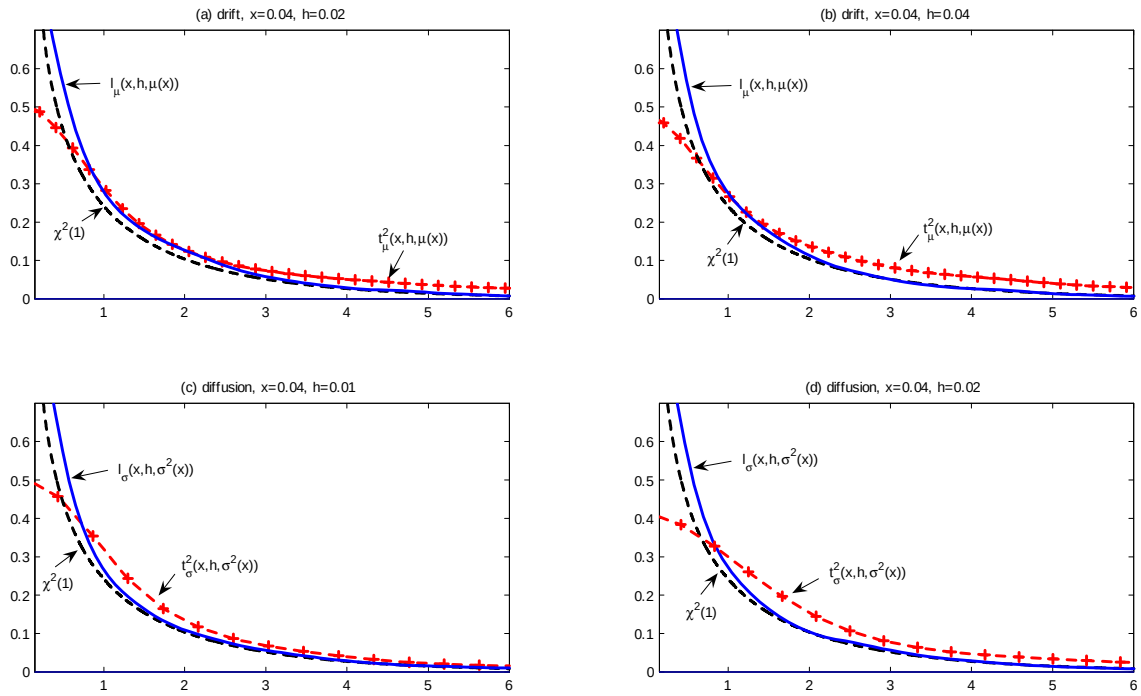

Figure 1: Density functions of the log EL ratios (solid line) and the squared $t-$statistics (dashed line marked with "$+$") evaluated at the true values for the drift ((a) and (b) with two different bandwidths) and diffusion ((c) and (d) with two different bandwidths) functions, based on 5000 replications. They are compared with their common limiting $\chi^2(1)$ density function (dashed line). DGP: $dX_t = 0.214592(0.085711 - X_t)dt + 0.07830X_t^{0.5}dB_t$. Sampling freqency $1/\Delta = 252$, time span $T = 30$, and the state point considered $x = 0.04$.

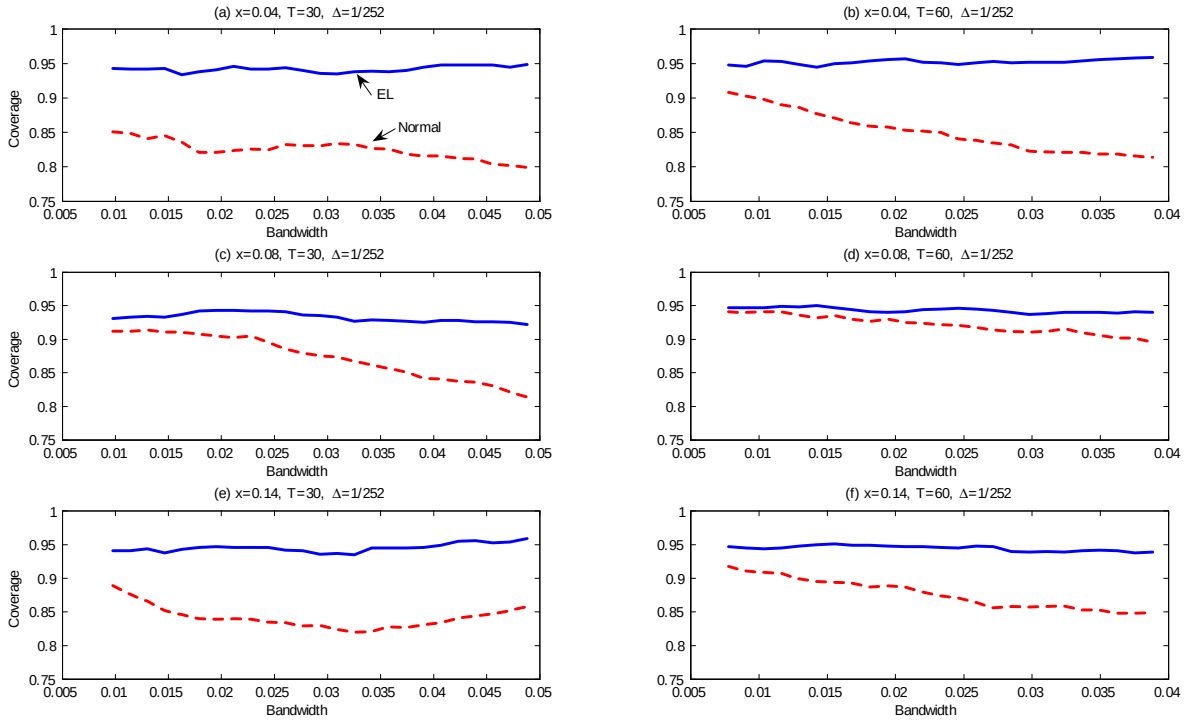

Figure 2: Empirical coverage rates of 95% pointwise normal and EL confidence intervals for the *drift* function. DGP: $dX_t = 0.214592(0.085711 - X_t)dt + 0.07830X_t^{0.5}dB_t$. Sampling freqency $1/\Delta = 252$ and time span $T = 30, 60$. (a) and (b): $x = 0.04$; (c) and (d): $x = 0.08$; (e) and (f): $x = 0.14$.

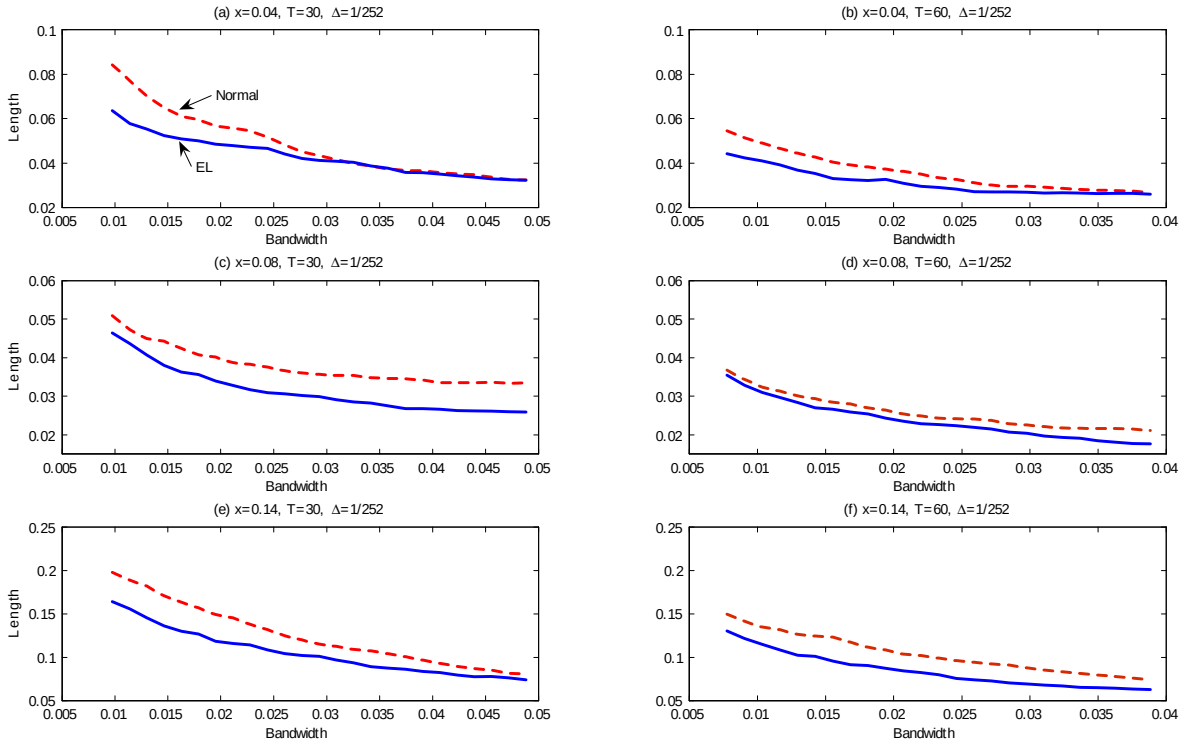

Figure 3: Adjusted lengths of 95% pointwise normal and EL confidence intervals for the *drift* function. DGP: $dX_t = 0.214592(0.085711 - X_t)dt + 0.07830X_t^{0.5}dB_t$. Sampling frequency $1/\Delta = 252$ and time span $T = 30, 60$. (a) and (b): $x = 0.04$; (c) and (d): $x = 0.08$; (e) and (f): $x = 0.14$.

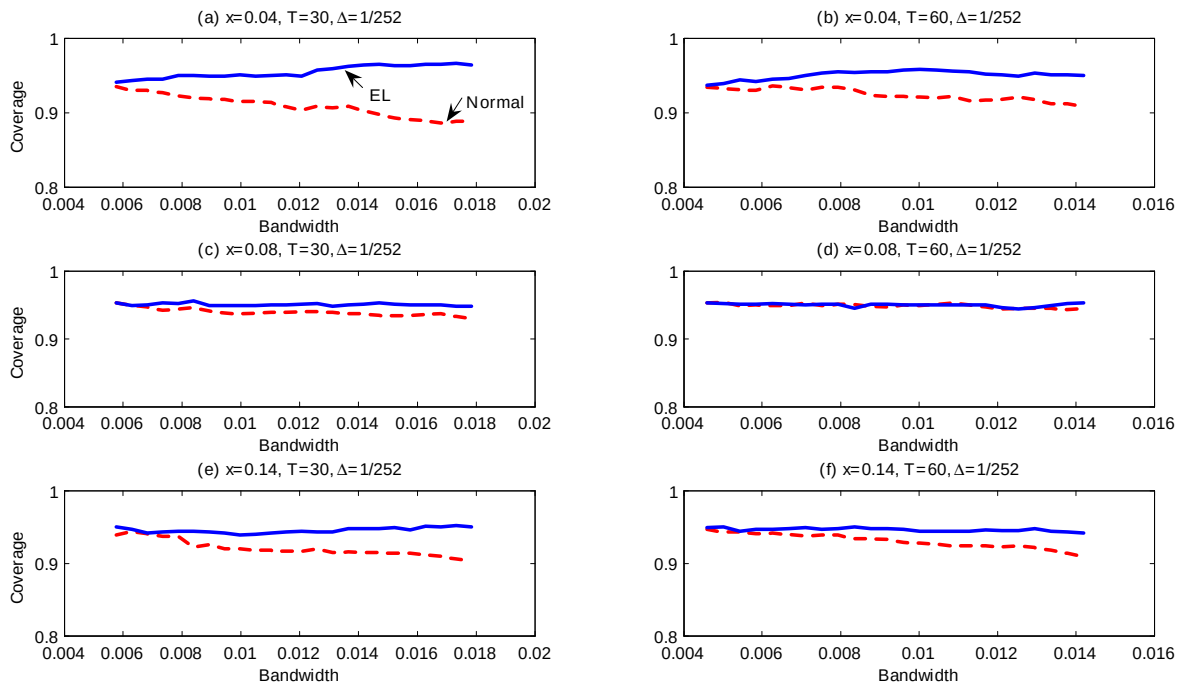

Figure 4: Empirical coverage rates of 95% point-wise normal and EL confidence intervals for the *diffusion* function. DGP: $dX_t = 0.214592(0.085711 - X_t)dt + 0.07830X_t^{0.5}dB_t$. Sampling frequency $1/\Delta = 252$ and time span $T = 30, 60$. (a) and (b): $x = 0.04$; (c) and (d): $x = 0.08$; (e) and (f): $x = 0.14$.

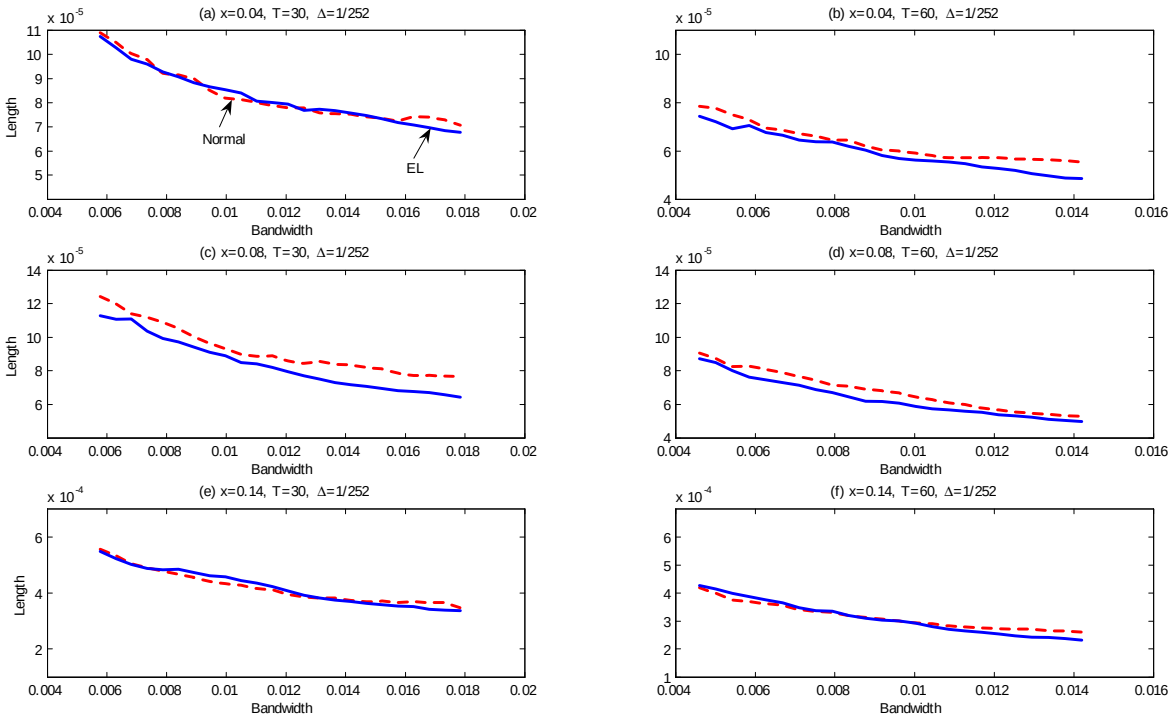

Figure 5: Adjusted lengths of 95% point-wise normal and EL confidence intervals for the *diffusion* function. DGP: $dX_t = 0.214592(0.085711 - X_t)dt + 0.07830X_t^{0.5}dB_t$. Sampling frequency $1/\Delta = 252$ and time span $T = 30, 60$. (a) and (b): $x = 0.04$; (c) and (d): $x = 0.08$; (e) and (f): $x = 0.14$.

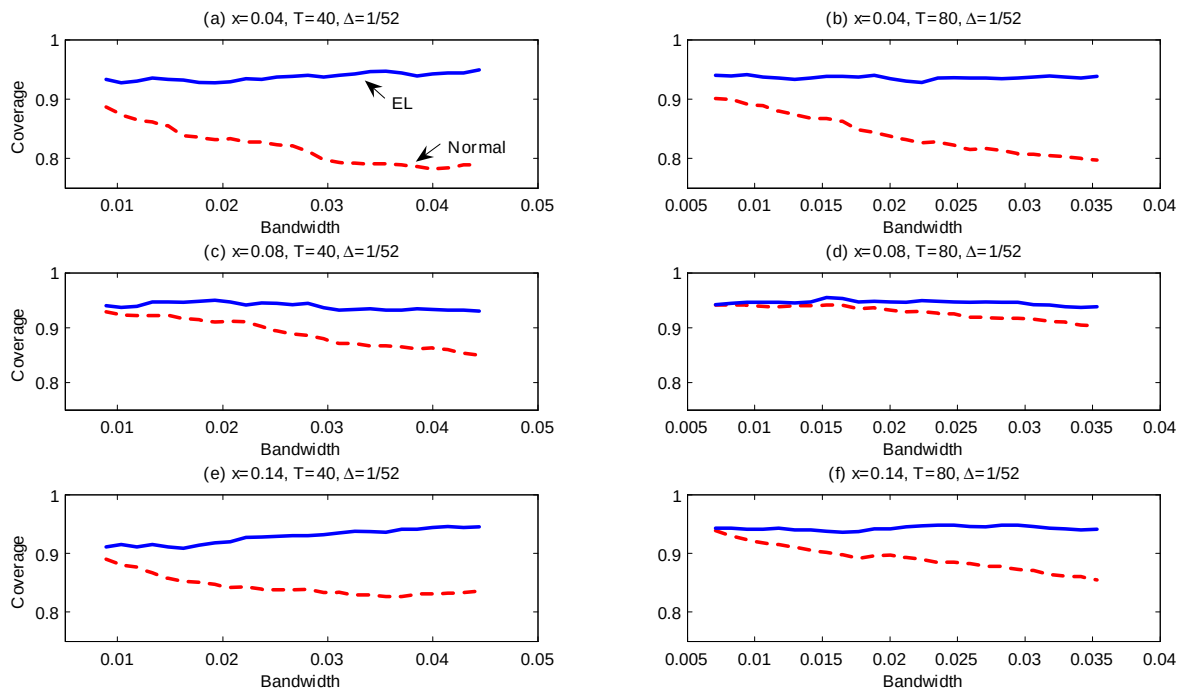

Figure 6: Empirical coverage rates of 95% pointwise normal and EL confidence intervals for the *drift* function. DGP: $dX_t = 0.214592(0.085711 - X_t)dt + 0.07830X_t^{0.5}dB_t$. Sampling frequency $1/\Delta = 52$ and time span $T = 40, 80$. (a) and (b): $x = 0.04$; (c) and (d): $x = 0.08$; (e) and (f): $x = 0.14$.

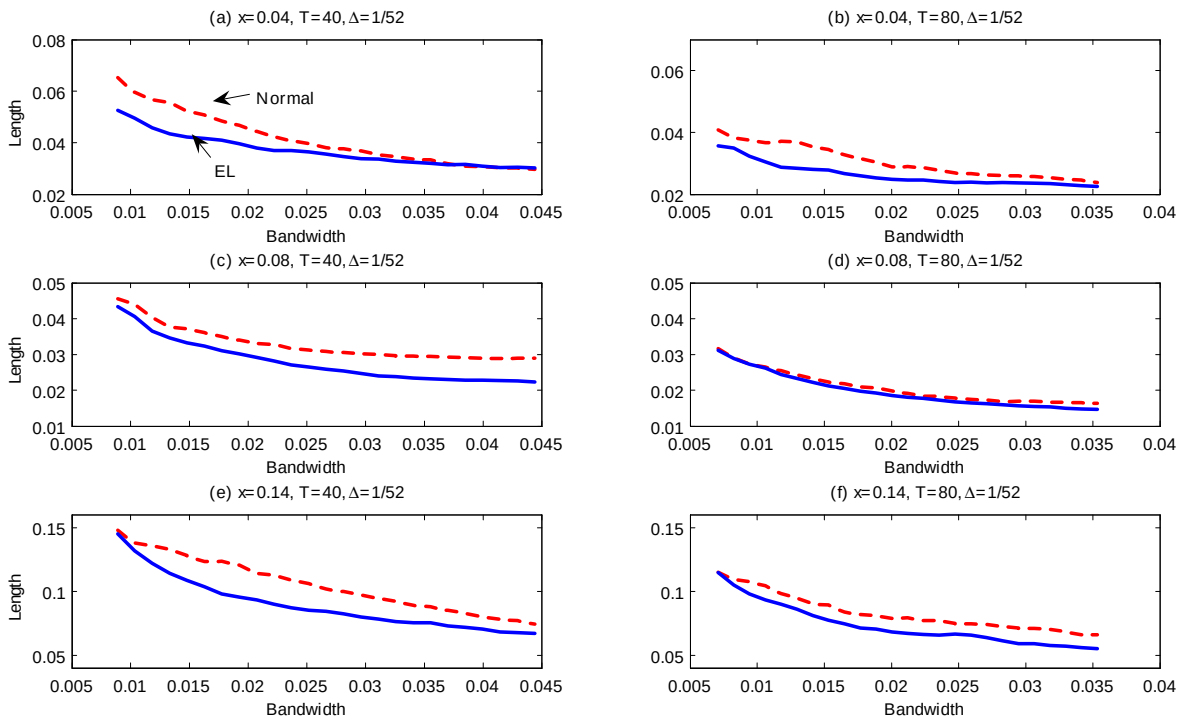

Figure 7: Adjusted lengths of 95% point-wise normal and EL confidence intervals for the *drift* function. DGP: $dX_t = 0.214592(0.085711 - X_t)dt + 0.07830X_t^{0.5}dB_t$. Sampling frequency $1/\Delta = 52$ and time span $T = 40, 80$. (a) and (b): $x = 0.04$; (c) and (d): $x = 0.08$; (e) and (f): $x = 0.14$.

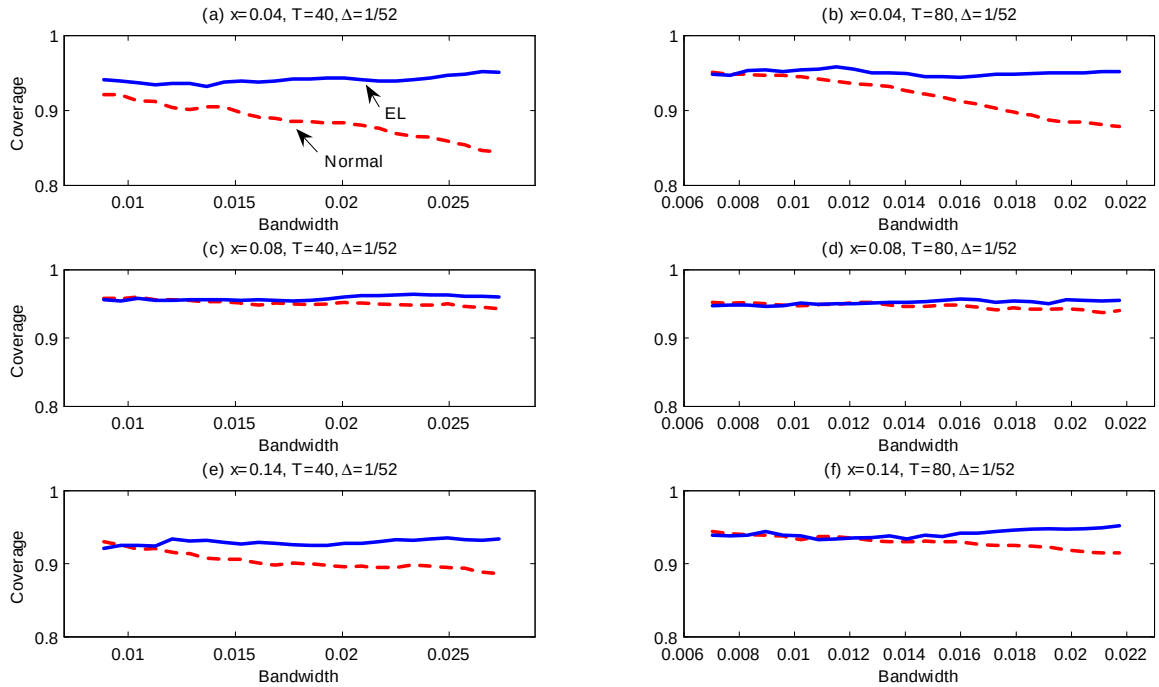

Figure 8: Empirical coverage rates of 95% point-wise normal and EL confidence intervals for the *diffusion* function. DGP: $dX_t = 0.214592(0.085711 - X_t)dt + 0.07830X_t^{0.5}dB_t$. Sampling frequency $1/\Delta = 52$ and time span $T = 40, 80$. (a) and (b): $x = 0.04$; (c) and (d): $x = 0.08$; (e) and (f): $x = 0.14$.

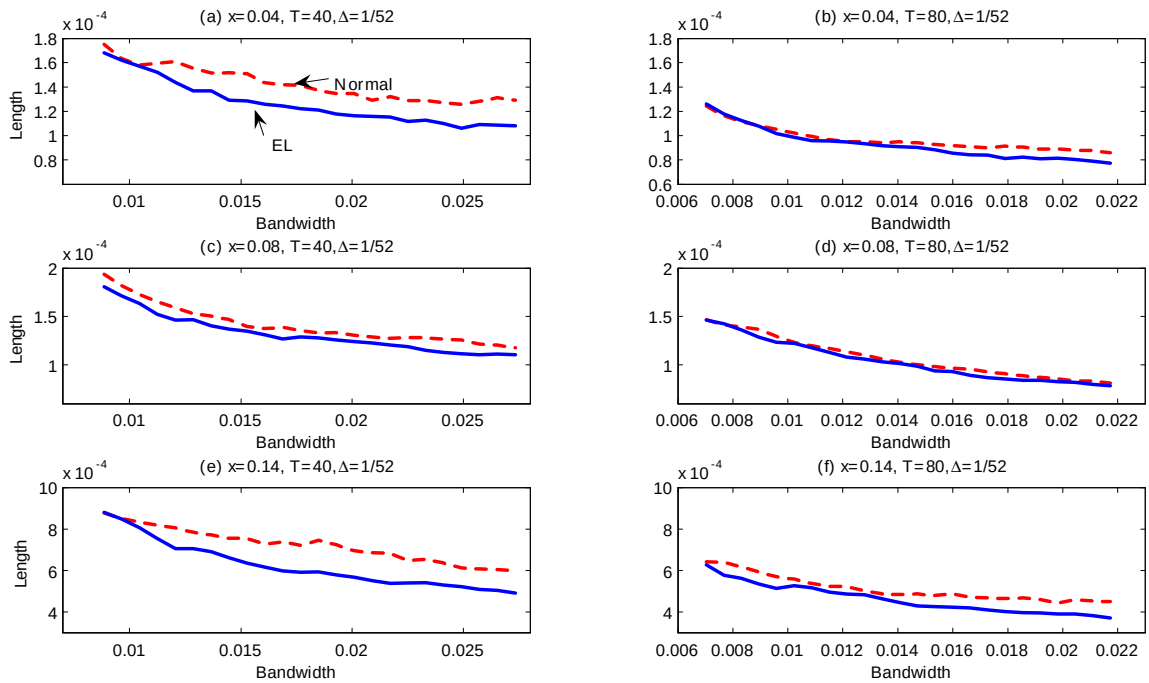

Figure 9: Adjusted lengths of 95% point-wise normal and EL confidence intervals for the *diffusion* function. DGP: $dX_t = 0.214592(0.085711 - X_t)dt + 0.07830X_t^{0.5}dB_t$. Sampling frequency $1/\Delta = 52$ and time span $T = 40, 80$. (a) and (b): $x = 0.04$; (c) and (d): $x = 0.08$; (e) and (f): $x = 0.14$.